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AT82.05 Artificial Intelligence: Natural Language Understanding (NLU)

A5: Optimization Human Preference & LLM-as-a-Judge

This assignment focuses on two critical aspects of modern Large Language Model (LLM) development: **Alignment** and **Evaluation**. You will leverage the Direct Preference Optimization (DPO) trainer to align a pre-trained model to be "Truthful" (avoiding hallucinations). Furthermore, you will learn how to build a reliable **LLM-as-a-Judge** pipeline to evaluate factuality.

Note: You are ENCOURAGED to work with your friends, but DISCOURAGED to blindly copy others' work. Both parties will be given 0.

Note: Comments should be provided sufficiently so we know you understand. Failure to do so can raise suspicion of possible copying/plagiarism.

Note: You will be graded upon (1) documentation, (2) experiment, (3) implementation.

Note: This is a one-week assignment, but start early.

Deliverables: The GitHub link containing the Jupyter notebook and a README.md of the GitHub repository.

Task 1. Dataset Preparation (0.5 point)

We will use a small, focused dataset designed to teach models to be truthful and avoid hallucinations.

Dataset: `jondurbin/truthy-dpo-v0.1`[1]

Task:

- Load the dataset using the Hugging Face `datasets` library.
- This dataset contains `prompt`, `chosen` (factual answer), and `rejected` (hallucinated/wrong answer).

Task 2. Training a Model with DPOTrainer

1) Implement the Direct Preference Optimization (DPO) training method with **DPOTrainer** using a pre-trained transformer model (e.g., `Qwen/Qwen2.5-1.5B-Instruct`) and fine-tune it using the training set from Task 1. (1 point)
2) Experiment with hyperparameters and report training performance (loss curves). (1 point)

HINT: Refer to the Hugging Face documentation for **DPOTrainer** implementation.[2].

Task 3. Pushing the Model to Hugging Face Hub (0.5 point)

1) Save the trained model and upload it to the Hugging Face Model Hub.
2) Provide a link to your uploaded model in your documentation.

Task 4. Evaluation: LLM-as-a-Judge with AlpacaEval (2 points)

To properly evaluate if your DPO training improved the model, we will use the industry-standard **AlpacaEval** benchmark dataset. You will build a pipeline where a strong LLM (e.g., GPT-4o-mini, Gemini) acts as an automatic judge to compare your models side-by-side.

1) **Step 1: Load AlpacaEval & Generate Responses**
   - Load the evaluation dataset: `tatsu-lab/alpaca_eval`[3].
   - **Note**: To bypass Hugging Face script restrictions, use the following code to load the raw JSON directly and filter for the `helpful_base` subset:

```python
from datasets import load_dataset

# Define the direct URL to the raw JSON file on the Hugging Face hub
data_url = "https://huggingface.co/datasets/tatsu-lab/alpaca_eval/resolve/main/alpaca_eval.json"

```

[1] https://huggingface.co/datasets/jondurbin/truthy-dpo-v0.1
[2] https://huggingface.co/docs/trl/main/dpo_trainer
[3] https://huggingface.co/datasets/tatsu-lab/alpaca_eval

```python
# Load the dataset using the 'json' builder, completely bypassing the disabled script
dataset = load_dataset("json", data_files=data_url)

# Filter the dataset to only include the 'helpful_base' subset
helpful_base_dataset = dataset['train'].filter(lambda x: x['dataset'] == 'helpful_base')
```

- Select a random sample of **15 prompts** from your filtered `helpful_base_dataset`.
- Generate answers for these 15 prompts using **BOTH** your pre-trained Base Model and your fine-tuned DPO Model. Keep track of which answer belongs to which model.

2) **Step 2: The Judge Prompt** Construct a prompt for your LLM Judge. We will instruct the Judge to output ONLY the name of the winner to make parsing extremely simple. Use this exact template:

   "You are a highly qualified and impartial judge evaluating two AI models. Your task is to determine which model provides a better, more accurate, and more helpful response to the user's instruction.

   ***User Instruction**: {instruction}*

   ***Model A (Base Model)**: {base_model_answer}*
   ***Model B (DPO Model)**: {dpo_model_answer}*

   Evaluate both models. Output ONLY your final verdict as exactly one of the following options, with no extra text or explanation: "Model A", "Model B", or "Tie"."

3) **Step 3: Evaluate and Collect Results** Iterate through your 15 test samples, send the prompts to your Judge LLM, and collect the verdicts. Your final output should look similar to the sample table below:

Sample ID	Instruction (Truncated)	Winner (Judge)
1	What are the primary differences between...	Model B
2	Write a python script to calculate the...	Model A
3	Explain the theory of relativity in simple...	Tie
4	Give me a list of 5 healthy breakfast...	Model B

Table 1. Sample format for storing your evaluation results

4) **Step 4: Calculate Win Rate** Count the total number of wins for Model B (your DPO Model). Calculate the Win Rate using the following formula:

$$\text{Win Rate} = \frac{\text{Model B Wins} + (0.5 \times \text{Ties})}{\text{Total Valid Evaluations}} \times 100$$

Report your final Win Rate. Discuss the results: Did your DPO training successfully improve the model's performance on the AlpacaEval benchmark?

Have fun with Alignment & Evaluation :-)